

Vishnu Datta

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EDUCATION

Indian Institute of Information Technology, Sri City

Bachelors of Technology in Computer Science and Engineering

Aug. 2023 – May 2027

Sri City, India

EXPERIENCE

Soundverse AI

Apr 2026 – Present

Software Development Engineer Intern

Remote

- Contributed to the development of **Soundverse 3.0**, overhauling platform architecture, modernizing the web platform, re-engineering media tooling, and developing scalable backend services
- Built a **Media Stitching API** that translates user editing prompts into automated audio and video editing pipelines, enabling complex timeline composition, trimming, overlays, transitions, and media transformations using **FFmpeg**
- Architected a scalable transcription and speaker diarization API, extending **Whisper** processing from **25MB** to **500MB+** audio inputs using silence-aware chunking, downsampling, and audio merging pipelines
- Currently engineering the backend infrastructure for **Music Distribution**, building scalable **APIs** and orchestration services for automated release validation, metadata processing, and digital music distribution workflows
- Engineered a scalable **Media Conversion API** supporting audio, video, and image conversions with advanced transcoding, compression, codec customization, and optimization using **FFmpeg** and **Pillow**
- Designed webhook-driven automation services that intelligently routed and categorized **Zammad** support tickets into dedicated **Slack** channels, improving internal operational efficiency

PROJECTS

CodeRep | *Next.js, TypeScript, PostgreSQL, Prisma, Clerk, Chrome Extension, Tailwind CSS* | GitHub 🐙 | [Live Demo](#)

- Engineered a full-stack web application with a **Chrome Extension (Manifest V3)** implementing spaced repetition algorithms for LeetCode revision, automating review scheduling for **20+ Daily Active Users**.
- Designed RESTful APIs using **Next.js App Router** and **Prisma ORM**, implementing dual authentication (**Clerk OAuth + token-based auth**) for reliable cross-platform synchronization.
- Developed a **MutationObserver-powered** content script to detect LeetCode submissions in real time, achieving **99% detection accuracy** through intelligent submission-state filtering.
- Built an adaptive spaced-repetition scheduling system using **exponential backoff multipliers (1.2x–4x)** to dynamically reduce unnecessary review frequency while reinforcing long-term problem retention.

FluenSee | *React, Node.js, MongoDB, Google Gemini API, Socket.IO, Stream SDK, JWT* | GitHub 🐙 | [Live Demo](#)

- Developed an AI-powered language tutor using **Google Gemini 2.5 Flash**, enabling real-time conversational practice with instant grammar correction through interactive role-play based learning sessions
- Integrated high-quality **1v1 video calling** using the **Stream Video SDK** and **Socket.IO**, enabling seamless peer-to-peer language exchange with synchronized real-time communication
- Built a scalable full-stack architecture using **React + Vite, Node.js**, and **MongoDB**, implementing secure authentication with **JWT** and HTTP-only cookie-based session handling

AlgoClash (In Progress) | *React, Node.js, Socket.IO, MongoDB, Piston API* | GitHub 🐙

- Developing a real-time **1v1 competitive coding platform** using **Socket.IO**, enabling synchronized multiplayer coding battles with isolated match state management and low-latency event synchronization
- Designed an automated tournament system generating scalable bracket structures for **32+ participants**, supporting dynamic round advancement and real-time match progression
- Built a secure remote code execution pipeline using the **Piston API**, supporting compilation and execution workflows across **8 programming languages** with standardized runtime handling

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript (ES6+), SQL, C, HTML, CSS

Frontend: React.js, Next.js, Tailwind CSS, Shadcn UI, Vite

Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, WebSockets, Socket.IO, Prisma ORM, JWT, OAuth, Clerk, Webhooks, Async Programming, Chrome Extension APIs

Databases: PostgreSQL, MongoDB, Redis, Supabase, Firebase

Developer Tools: Git, GitHub, Docker, Postman, FFmpeg, Pillow, Azure